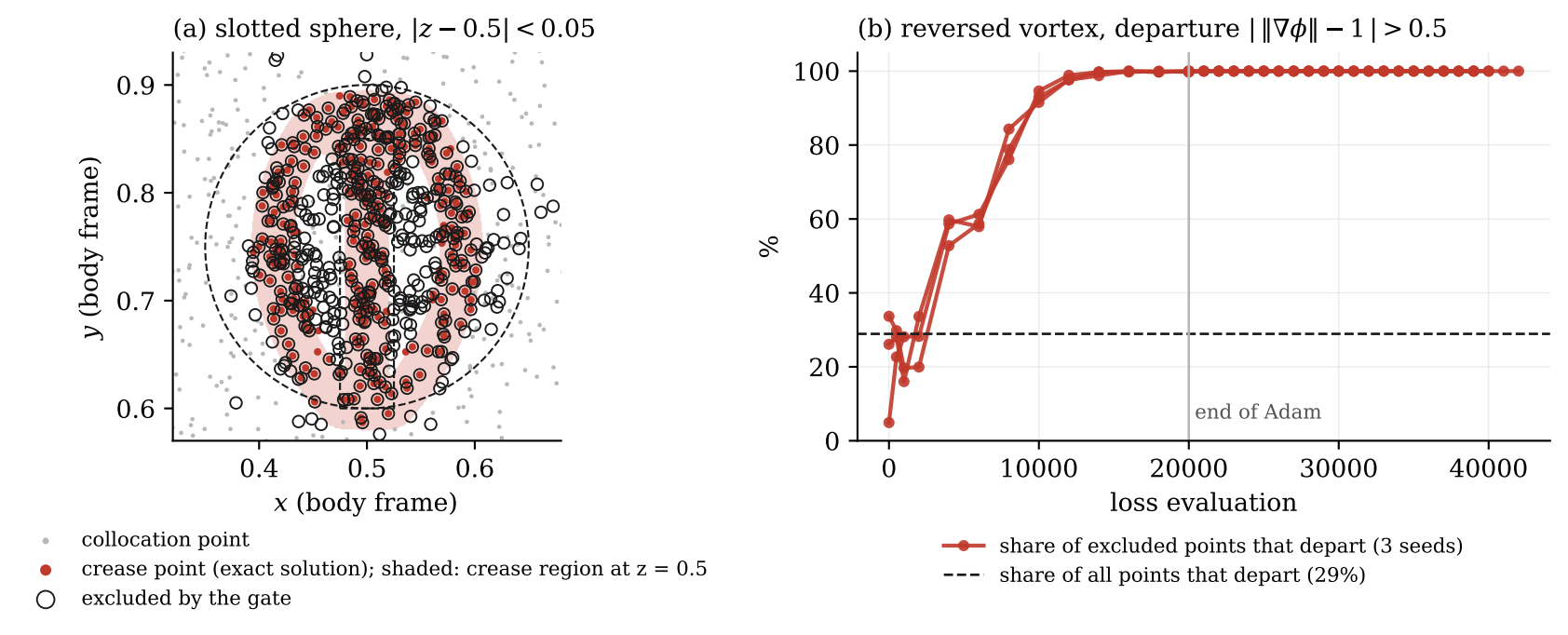

(a) slotted sphere, $|z - 0.5| < 0.05$
0.9
0.8
0.7
0.6
y (body frame)
0.4
0.5
0.6
x (body frame)
collocation point
crease point (exact solution); shaded: crease region at z = 0.5
excluded by the gate
(b) reversed vortex, departure $|\,\|\nabla\phi\| - 1\,| > 0.5$
100
80
60
40
20
0
%
0
10000
20000
30000
40000
loss evaluation
end of Adam
share of excluded points that depart (3 seeds)
share of all points that depart (29%)